GEp-4 Reboot Feasibility Study

Data anchors, present assessment, and critical-path questions

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Working configuration: 10.7 GeV beam, 75 μA, 30 cm LH_2, positive-polarity HMS proton arm, and NPS electron arm near 3 m.

What we set out to establish

Physics reach

Does the HMS–NPS configuration provide enough accepted elastic events at $Q^2 \simeq 11\ \mathrm{GeV}^2$ for a competitive form-factor measurement?

NPS response and backgrounds

Can a high NPS threshold retain elastic efficiency while controlling electron, photon, and charged-hadron backgrounds?

Trigger and DAQ

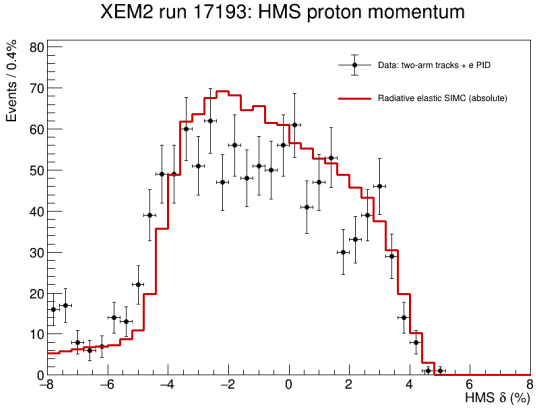
What are the real HMS and NPS singles rates, the accidental-coincidence rate, and the required Hall C DAQ capability?

Experimental design

Are the NPS distance, FPP configuration, HMS trigger logic, and coincidence timing mutually compatible?

Approach: anchor SIMC and GEANT4 to real Hall C data before extrapolating.

Elastic yield: a close HMS benchmark supports the proposal



XEM2 run 17193: HMS $+6.476$ GeV/c at 19.52°; proposed HMS $+6.735$ GeV/c at 18.80°.

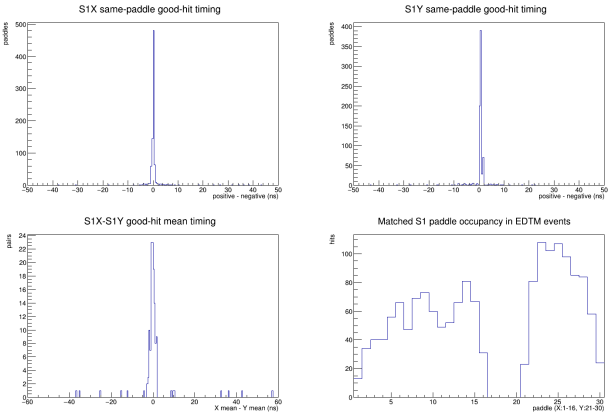
- Broad HMS acceptance: data/SIMC $= 0.936$.
- Tight elastic selection: $800/1034.34 = 0.7734$.
- At fixed 30 cm target length, NPS at 3 m gains a factor 4.11 over SHMS-like electron acceptance.
- Transfer from the XEM2 setting to $Q^2 \simeq 11\ \mathrm{GeV}^2$: SIMC ratio $= 0.8865$.
- Direct $Q^2 \simeq 11\ \mathrm{GeV}^2$ target scan: 20/10 cm $= 1.85$ and 30/10 cm $= 2.57$; the final 10 cm supplies 28% of the 30 cm yield.

45 PAC-day projection

Before NPS response	**10.69 M**
70% threshold + fiducial	**7.70 M**
80% threshold + fiducial	**7.51 M**

Response values remain provisional (3×3 energy, radiative photons, passive material). The sweeper will be absent in GEp; retaining it here is a small conservative effect.

HMS trigger: the practical S1 rate is 40–60 kHz



Run 17193 unbiased EDTM windows. A valid plane hit requires both PMTs on one physical paddle; valid X and Y paddle numbers are independent.

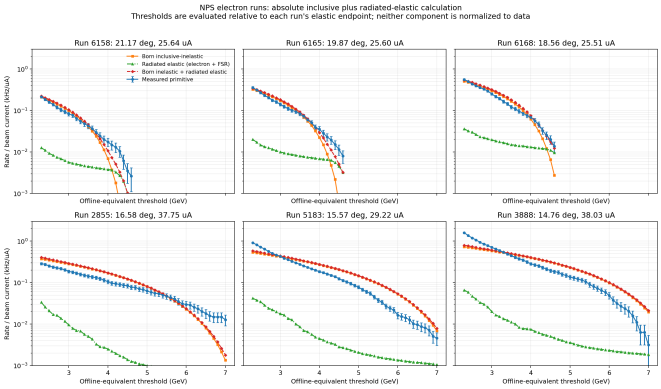
- The branch previously labeled "S1X AND S1Y" is actually **S1X OR S1Y**; its GEp-scaled rate is about 810 kHz.
- EDTM plane rates reproduce direct scalers to 3–4%.
- Requiring a double-ended paddle hit in both planes gives 12.73–14.84 kHz at the run conditions.
- Naive GEp scaling gives 51–60 kHz; scaler inclusion–exclusion gives 39.9 kHz.
- HMS 3-of-4 scales to about **2.9 kHz** if it is compatible with the FPP configuration.

Accidental scale

For 50 kHz HMS and 1 MHz NPS:

$$R_{\mathrm{acc}} = R_H R_N \Delta t = \begin{cases} 1\ \mathrm{kHz}, & 20\ \mathrm{ns} \\ 2\ \mathrm{kHz}, & 40\ \mathrm{ns}. \end{cases}$$

NPS singles: the inclusive-electron component is data anchored



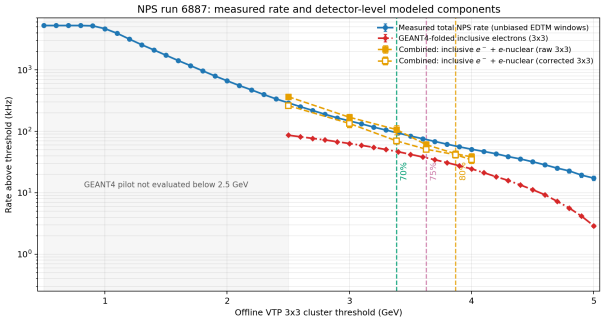
- Three-pass NPS electron data: generator/data $= 0.95$–1.35.
- Five-pass NPS electron data: trend reproduced; most discrepancies are conservative overpredictions.
- XEM2 SHMS electron-enriched trigger: 447.7 Hz predicted versus 592 Hz measured, or 74.8%.

GEp full-face prediction

Threshold	Inclusive e^-	+ radiated elastic
70%	14.34 kHz	**14.42 kHz**
75%	10.26 kHz	**10.30 kHz**
80%	7.05 kHz	**7.08 kHz**

These are genuine-electron rates, not the total NPS trigger rate.

Run 6887 establishes a large non-electron excess

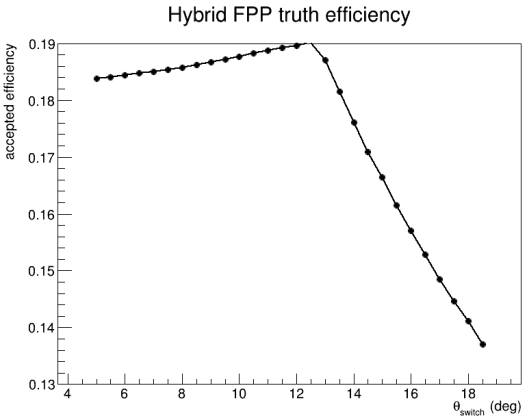


Run 6887: DVCS production, NPS on the photon arm; 8.456 GeV beam, 29.4 μA, 10 cm LH_2; NPS at 16.57° and 3.57 m. VTP single-cluster trigger/readout settings: 0.50/0.25 GeV; 8,972 unbiased 1-μs EDTM windows after the 25–35 μA current selection.

What the data establish

At 70–80%, inclusive electrons alone explain only 25–50% of the rate. Adding importance-sampled electronuclear production gives 107, 61, and 43 kHz with raw 3x3 energy, or 70, 51, and 42 kHz with the inherited correction. The six direct points use the 0.25-GeV production cut. **The background is now at the required scale**; the high-threshold deficit and 3x3 energy convention remain open.

A single analyzer preserves the small-angle FPP acceptance



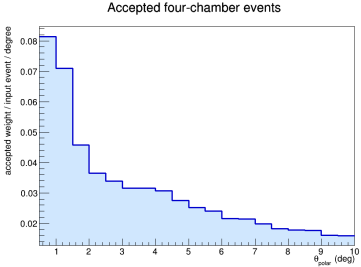
- Legacy FPP frame: one 22-inch CH_2 analyzer.
- $\theta_{\mathrm{polar}} < 12^{\circ}$: reconstruct with FPP1–FPP4; four chambers constrain the left/right ambiguity.
- $\theta_{\mathrm{polar}} > 12^{\circ}$: reconstruct with FPP1 and FPP2; the larger deflection resolves the ambiguity and avoids downstream cone-test loss.
- Truth efficiency peaks near the 12° transition.

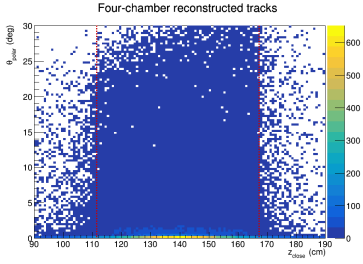
Four-chamber closure

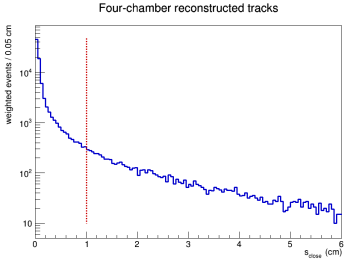
All 59,113 truth-selected events from 0° to 10° reconstruct in the ideal fit, with no large angular outliers.

Ideal closure: truth-associated hits and exact drifts; efficiency, resolution, noise, and pattern recognition remain.

Hall C chamber geometry and wire-level tracking







- Ported the four measured chamber positions and twelve U/X/V planes from the GEp-III parameters: 104/83/104 wires per chamber with 2 cm pitch.
- Digitize plane and wire number plus unsigned drift distance; enumerate the left/right signs and retain the minimum reduced-χ^2 $x(z), y(z)$ fit.
- Calculate the FPP observables relative to the HMS track and apply the cone test. The efficiency density at left includes the analyzer-volume, cone-test and 1 cm s_{close} requirements on truth-associated candidates.

Ideal closure: truth-associated hits and exact drifts; the fit receives neither the truth coordinate nor its sign. Detector response and multi-track recognition remain.

Single-track vs. Multi-track FOM

Use the same beam-polarization-weighted normalization per incident proton:

$$\mathrm{FOM} = P_e^2\,\epsilon_{\mathrm{FPP}}\langle A_y^2\rangle, \qquad P_e = 0.85.$$

Estimate / selection	Useful efficiency	A_y input	FOM / proton
Andrew's planning estimate	20% assumed	$\sim$0.050	$\sim 3.60 \times 10^{-4}$
Hall C: single tracks (no 1° cut)	12.492%	0.05859 RMS	3.10×10^{-4}
Hall C: smallest angle $> 1^\circ$	25.892%	0.06704 RMS	8.41×10^{-4}*

What explains the initial discrepancy

Our original 7.56×10^{-4} included multi-track events. Restricting to single tracks gives a result only **14% below Andrew's** 3.6×10^{-4} estimate, with the GEp-III A_y parametrization.

Andrew uses an estimated average A_y; the simulation integrates A_y^2 event by event. FOM agreement does not imply identical efficiency or cuts.

*Multi-track FOM is provisional: it assumes the single-track A_y model applies.

Analyzing-power parametrization and FOM weighting

The same construction appears in the SBS planning code and Andrew's description:

$$A_y(p, p_T) = \left(0.0342 + \frac{0.348}{p}\right)\sqrt{2eb}\, p_T e^{-bp_T^2}, \qquad b = 2.42614689.$$

b: inverse-variance-weighted mean of the three high-Q^2 fits; $A_y^{\max} = 0.0342 + 0.348/p$. Use $p_T = p\sin\theta$, with p, p_T in GeV/c.

Same exclusive single-track sample: $p = 6.7$ GeV/c, $\epsilon = 12.492\%$, $P_e = 0.85$.

Averaging convention	Effective A_y	FOM / incident proton
Square of mean: $P_e^2\epsilon\langle A_y\rangle^2$	0.05484	2.714×10^{-4}
Mean square: $P_e^2\epsilon\langle A_y^2\rangle$	0.05859 RMS	3.099×10^{-4}

Event-by-event weighting gives 14.2% more FOM, not a factor of two. No parametrization mismatch was found.

Table VII's $\bar{A}_y$ is an asymmetry-calibrated effective average, not an RMS. Its $0.06 \le p_T \le 1.2$ GeV/c range also differs from our $0.07 \le p_T \le 1.0$ GeV/c selection (no explicit 1° cut here).

SBS asymmetry supports the analyzing-power scale

Andrew's preliminary $Q^2 = 11$ GeV^2 data give a sine amplitude $|s| = 0.02971 \pm 0.00403$. In the leading dipole approximation,

$$A_{y,\mathrm{eff}} \simeq \frac{|s|}{P_e P_\ell |\sin\chi|}, \qquad P_e \simeq 0.87.$$

Illustrative Born kinematics: $E = 10.7$ GeV, $Q^2 = 11$ GeV^2, $\mu_p G_E/G_M = 0.2$ give $\epsilon = 0.684$, $P_\ell = 0.729$, $P_t = -0.027$ for a fully polarized beam.

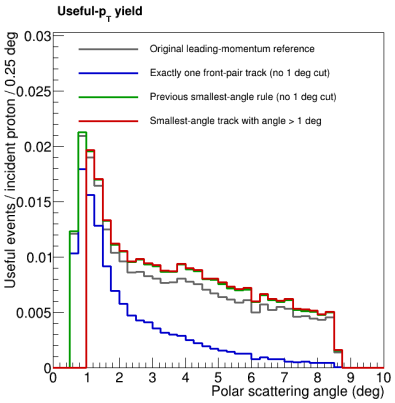
Estimate / assumption	Effective average A_y
SBS data, $\lvert\sin\chi\rvert = 1$	0.0468 ± 0.0064
SBS data, $\chi = 1.3$ rad	0.0486 ± 0.0066
Our simulated single-track mean	0.05484

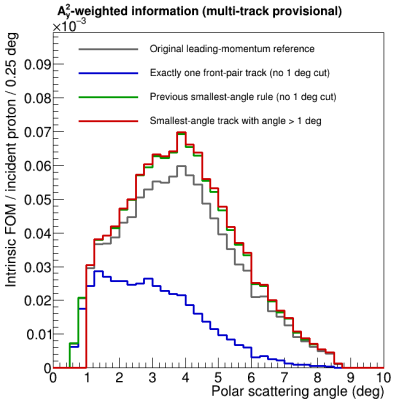
The estimates agree within roughly one statistical standard deviation. Varying $\mu_p G_E/G_M$ from 0 to 0.4 changes P_ℓ by less than 0.5%.

Errors shown propagate the plotted asymmetry uncertainty only. This estimate uses approximate central kinematics and spin precession; acceptance averaging and background dilution are not determined here. It estimates a mean A_y, not the RMS, and does not validate multi-track FOM.

Source: A. Puckett, SBS Collaboration Meeting, 2 March 2026, slides 9, 10, 36.

Smallest-angle selection recovers useful multi-track events





Rule: choose the smallest polar angle strictly above 1°, then apply the existing chamber, cone, closest-approach, and useful-p_T cuts.

Result: 25,892 useful events, including 16,229 multi-track events, per 100,000 protons; 98.79% of the previous FOM retained.

Multi-track recovery could improve the FPP statistical reach

The single-track calculation supports Andrew's FOM scale. The simulation suggests that retaining some multi-track events could do better.

$$\sigma_{\text{component}} = \sqrt{\frac{2}{N\,\text{FOM}}}, \qquad N = 7.5\text{–}7.7 \times 10^{6}.$$

FPP selection	FOM / incident proton	$\sigma_{\text{component}}$
Exclusive single tracks	3.10×10^{-4}	0.0293–0.0290
Smallest angle $> 1^{\circ}$, including multi-track	8.41×10^{-4}	0.0178–0.0176*

*Provisional: GEp-III single-track A_y applied to ideal, truth-associated multi-track events. Detector efficiency, resolution, noise, and multi-track pattern recognition remain to be included. These are polarization-component uncertainties, not uncertainties in $\mu_p G_E / G_M$.